WORKED SOLUTION

STEP II 2016, Question 1

Pure • Parametric Curves and Tangents

> **STEP II 2016, Question 1**
>
> Read the question before you go on. It is in the STEP Support Programme's question database:
>
> step.maths.org/questions/16-s2-q1

The key idea

Work with the parameters rather than the coordinates. The right angle at O turns into a single, very simple relation between the parameters p and q of the two points, and the tangent at a general point has a neat equation in t. Once the two tangents are intersected, the coordinates of the meeting point come out as symmetric functions of p and q, and the relation between p and q is exactly what is needed to eliminate them.

Solution

The tangent at a general point

At the point with parameter t,

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{\mathrm{d}y/\mathrm{d}t}{\mathrm{d}x/\mathrm{d}t} = \frac{3t^2}{2t} = \frac{3t}{2} \qquad (t \neq 0).$$

So the tangent at (t^2, t^3) is $y - t^3 = \frac{3t}{2}(x - t^2)$, that is,

$$2y = 3tx - t^3. \tag{$*$}$$

(At $t = 0$ this gives $y = 0$, which is the tangent at the cusp, but we shall not need that point.)

The right angle

Let $P = (p^2, p^3)$ and $Q = (q^2, q^3)$. For $\angle POQ$ to make sense, neither point is O, so $p \neq 0$ and $q \neq 0$. The angle is a right angle exactly when $\overrightarrow{OP} \cdot \overrightarrow{OQ} = 0$:

$$p^2q^2 + p^3q^3 = p^2q^2(1 + pq) = 0.$$

Since $p^2q^2 \neq 0$, this forces

$$pq = -1.$$

In particular $p \neq q$ (their product is negative), so the tangents at P and Q have different gradients $\frac{3p}{2} \neq \frac{3q}{2}$ and really do meet in a single point.

Where the tangents meet

Subtracting the two versions of $(*)$, with $t = p$ and $t = q$:

$$3(p-q)x = p^3 - q^3 = (p-q)(p^2+pq+q^2).$$

Dividing by $p - q \neq 0$ and using $pq = -1$,

$$3x = p^2 + pq + q^2 = p^2 + q^2 - 1.$$

Substituting back into $(*)$ with $t = p$:

$$\begin{aligned} 2y &= p(p^2+pq+q^2) - p^3 \\ &= p^2q + pq^2 = pq(p+q) = -(p+q). \end{aligned}$$

Ask yourself We have x and y in terms of two parameters. How do we get a single equation in x and y?

Square y: the cross term $2pq$ that appears is a known constant, and what remains is $p^2 + q^2$, which is exactly what $3x$ contains. Indeed,

$$4y^2 = (p+q)^2 = p^2 + q^2 + 2pq = p^2 + q^2 - 2 = (3x+1) - 2 = 3x - 1.$$

So the tangents at P and Q meet on C_2: $4y^2 = 3x - 1$, as required.

(In fact every point of C_2 arises: $2y = -(p+q) = \frac{1}{p} - p$ takes every real value as p runs through the positive reals, and x is then fixed by the equation of C_2.)

Do C_1 and C_2 meet?

Substitute $x = t^2$, $y = t^3$ into the equation of C_2:

$$4t^6 = 3t^2 - 1 \iff 4u^3 - 3u + 1 = 0, \quad \text{where } u = t^2 \geq 0.$$

Clearly $u = -1$ is a root, and dividing out the factor $u + 1$ gives

$$4u^3 - 3u + 1 = (u+1)(4u^2 - 4u + 1) = (u+1)(2u-1)^2.$$

Since $u \geq 0$, the only possibility is $u = \frac{1}{2}$, so $t = \pm\frac{1}{\sqrt{2}}$. Hence the curves *do* meet, at exactly two points

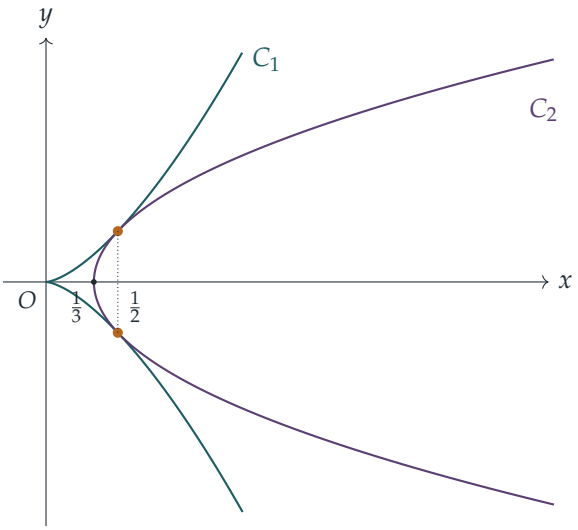
$$\left(\tfrac{1}{2},\ \tfrac{1}{2\sqrt{2}}\right) \quad \text{and} \quad \left(\tfrac{1}{2},\ -\tfrac{1}{2\sqrt{2}}\right).$$

The repeated root tells us more: the curves touch there rather than cross. We can confirm this directly. At $t = \frac{1}{\sqrt{2}}$ the gradient of C_1 is $\frac{3t}{2} = \frac{3}{2\sqrt{2}}$. Differentiating $4y^2 = 3x - 1$ implicitly gives $8y\frac{\mathrm{d}y}{\mathrm{d}x} = 3$, so at $y = \frac{1}{2\sqrt{2}}$ the gradient of C_2 is $\frac{3}{8y} = \frac{3}{2\sqrt{2}}$, the same. By symmetry in the x-axis the same holds at the other point.

Since $4u^3 - 3u + 1 = (u+1)(2u-1)^2 \geq 0$ for $u \geq 0$, the curve C_1 never goes inside the parabola: C_2 lies between the two branches of C_1 except at the two points of contact.

Sketch

C_1 satisfies $y^2 = x^3$: it lies in $x \geq 0$, is symmetric in the x-axis, and has a cusp at O (both branches have gradient $\frac{3t}{2} \to 0$, so they leave O along the positive x-axis). C_2 is the parabola $x = \frac{1}{3}(4y^2 + 1)$, with vertex $(\frac{1}{3}, 0)$, opening to the right and also symmetric in the x-axis. The parabola sits inside the cusp, between the two branches of C_1, and touches each branch once, at $x = \frac{1}{2}$.



Checks

- Take $p = 1$, $q = -1$: $P = (1, 1)$, $Q = (1, -1)$, and $\overrightarrow{OP} \cdot \overrightarrow{OQ} = 1 - 1 = 0$. The tangents are $2y = 3x - 1$ and $2y = -3x + 1$, meeting at $(\frac{1}{3}, 0)$, the vertex of C_2.
- Take $p = 2$, $q = -\frac{1}{2}$: the tangents $2y = 6x - 8$ and $2y = -\frac{3}{2}x + \frac{1}{8}$ meet at $x = \frac{13}{12}$, $y = -\frac{3}{4}$, and $4y^2 = \frac{9}{4} = 3 \cdot \frac{13}{12} - 1$.
- The point of contact $(\frac{1}{2}, \frac{1}{2\sqrt{2}}) \approx (0.5, 0.35355)$ satisfies $4y^2 = \frac{1}{2} = 3x - 1$, and both gradients there are ≈ 1.06066. A computer check of 10^5 random pairs with $pq = -1$ found every intersection on C_2 to within rounding error.

Common slips

- Forgetting to exclude $p = 0$ or $q = 0$: the factor p^2q^2 must be non-zero before dividing to get $pq = -1$.
- Dividing by $p - q$ without noting why it is non-zero.
- Stopping at "$4u^3 - 3u + 1 = 0$ has a root" without checking that it gives $u = t^2 \geq 0$; the root $u = -1$ does not correspond to any point of C_1.

Beyond STEP

The curve $y^2 = x^3$ is the classic *cuspidal cubic*, and the trick of meeting a curve with a parametrised curve and reading tangency from a repeated root is the starting point of algebraic geometry: intersection multiplicities count how many times curves meet, and Bézout's theorem says that a cubic and a conic meet in six points counted properly. Here the six are the double roots $t = \pm\frac{1}{\sqrt{2}}$ (four in all) and the complex pair $t = \pm i$. A friendly introduction is Kirwan, *Complex Algebraic Curves*.

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